Chem 210D: Modern Applications of Inorganic Chemistry

Prof. Charlie Cox

Fall Semester 2026

Lecture and Discussion Meeting Times

Component	Time	Location
Lecture	TuTh 11:45 am - 1:00 pm	Gross Hall 107
Disc. 10D	W 10:05 am - 11:20 am	LSRC A155
Disc. 11D	W 11:45 am - 1:00 pm	French 1243
Disc. 12D	W 4:40 pm - 5:55 pm	French 2237
Disc. 13D	W 1:25 pm - 2:40 pm	Old Chem 101
Disc. 14D	W 11:45 am - 1:00 pm	Bio Sci 130
Disc. 15D	W 3:05 pm - 4:20 pm	Physics 154
Disc. 16D	W 11:45 am - 1:00 pm	Bio Sci 063
Disc. 17D	W 10:05 am - 11:20 am	Gross 104
Disc. 18D	W 4:40 pm - 5:55 pm	Gross 105

Professor Information

Prof. Charlie Cox (he/him/his) ⇔ charlie.cox@duke.edu ⇔ French Family Science 1222

Office hours will be on Wednesday from 9:30 – 10:30 am and Tuesday from 4:30 - 5:30 pm – both in French 1218. These are Q & A. I will have problem-solving office hours on Monday from 4:45-5:45 pm in Physics 128. The problem-solving office hours are recorded.

Teaching Assistants

The teaching assistants or TAs are the true masterminds behind the course. I could not do this without them.

TA Name	**Email**	**Discussion Sections**	**Office Hours**
Taylor Thorsen	taylor.thorsen@duke.edu	10D, 11D, 13D	Thursday 5:00 pm-6:00 pm Problem Solving Physics 128
Heber Ford	Heber.ford@duke.edu	12D, 15D, 14D	Tuesday 5:30pm - 6:30pm & Thursday 1:30 pm - 2:30 pm both times are in French 1218
Ayan Jung	ayan.jung@duke.edu	16D, 17D	Tuesday 1:00 pm - 3:00 pm in French 1218
Mariam Dohadwala	mariam.dohadwala@duke.edu	18D	Monday 3:00 pm - 4:45 pm in French 1218

Learning Assistant

Jack Ringel ⇔ jack.ringel@duke.edu

Given the enrollment (> 200 students) in Chem 210, we will have a learning assistant who will provide additional support for the course through office hours, extra review sessions, and message board support.

Course Description

Chem 210D completes the introductory course sequence in chemistry by covering chemistry across the periodic table and introducing inorganic structure and reactivity. Specific concepts include nuclear reactivity, quantum mechanical structure, solids, molecular orbital theory for more complex molecules, solubility and solution chemistry, redox reactions, and coordination complexes.

Course Prerequisites

Chem 20, 21, 101, or 110 is a prerequisite for Chem 210. We will draw on concepts from introductory chemistry. I will provide screencasts that review some of the prerequisite content.

Course Materials

Tro, Structure and Properties, 3rd Edition with Mastering Chemistry, Required These materials can be purchased at the bookstore.

Course Structure

There will be lectures on Tu/Th from 11:45 am – 1:00 pm in Gross Hall 107. The discussion sections are held on Wednesday at 10:05, 11:45, 1:25, 3:05, or 4:40. Students must attend the discussion section in which they are enrolled to receive credit.

Approximate Schedule of Topics

Given that this is my first time teaching the course in a couple of years, I prefer not to provide a rigid schedule, as I aim to adjust the course pace to meet my students' needs. Every class is different. There is no continuation of Chem 210, and I have designed the schedule to cover the most important concepts, preparing us for future courses and standardized exams.

Topic	**# of Lectures**
Nuclear and Atomic Structure	5
Solutions and Solubility	4
Acid-Base Reactions	6
Coordination Complexes	5
Electrochemistry	4
Solid State Chemistry	2

Course Philosophy

Chemistry is challenging. I am not structuring the course with the intent that everyone will become a chemistry major, and I realize that most of you do not want to take it. However, my course aims to introduce concepts in an accessible way that highlights the wonders of chemistry and presents applications in medicine, materials, and sustainability. Students will grapple with some concepts, but we are here to support students in their journey in this course. Everyone in this class can succeed, and everyone should learn something. Apart from exams, I encourage students to work together! This class is not graded on a "curve" that rewards competition. Everyone can succeed. Therefore, working together benefits everyone.

Evaluation

Students are evaluated based on two midterm exams, discussion activities, Mastering Chemistry homework, and a comprehensive final exam. The better of the midterm exam average or the final exam will account for **50%**, and the lower of the two will account for **35%** of the final grade. The discussion scores and the Mastering Chemistry homework will account for the remaining **15%** of the final grade. Each discussion will be worth 5 points, with points stemming from arriving on time, participating in group activities, and not leaving early. Each Mastering Chemistry homework assignment will be worth 10 points. Students who earn 93% on homework and discussion points will receive full credit; a 92% will be scaled to 99%, and so on.

Exams

Exams are closed-note/closed-book. Pertinent equations and constants will be provided. Midterm exams are given in lecture on the following dates: **October 1** and **November 12**. The final exam is on **December 9th from 2:00 – 5:00 pm**. The registrar's office schedules final exams, and I am not permitted to change the exam date or administer the exam early. In the weeks following exams, there will be an opportunity to regain points through a resurrection quiz administered during the discussion section. To be eligible to take the resurrection quiz, students must correct their exams in accordance with the guidelines posted on Canvas. The total points gained depend on the percentage earned on the resurrection quiz: 100% corresponds to regaining 25% of the points lost on the original exam; 90%, to 22.5%; 80%, to 20%; and so on. Therefore, a student who earns a 90 on the original exam can earn up to 92.5; a student who earns an 80 can earn up to 85; a student who earns a 60 can earn up to 70. For exams with averages below 65%, the points regained will exceed 25%, with the exact amount depending on the exam average.

$$\textbf{Exam Score} = \textbf{Student Score} + (100 - \textbf{Student Score}) \times \left(\frac{\textbf{Points Earned}}{\textbf{Points Possible}}\right) \times 0.25$$

A STIN form is not acceptable documentation for missing an exam or a resurrection quiz.

During exams, the proctors and I do not answer questions because not every enrolled student is taking the exam in Gross Hall 107. Some are taking the exam in the testing center or later.

Final Grades

Final grades will be assigned using an **80-70-60-50** scale for **A-D**, with the upper 25% and lower 25% of each range representing a **+** or **-** mark. Therefore, a 95-100 is an A+, 85-94 is an A, and 80-84 is an A-. Once final grades are assigned, I cannot change them unless there is an error or something was overlooked. Once the grades are in DukeHub, they are final.

Regrade Requests

Please submit regrade requests through the appropriate channel. We will provide a form for students to enter their requests. The entire exam may be regraded at the graders' discretion. Therefore, you may receive a lower score after the grade. The regrade score is final. Meeting with the instructional team will not change a grade. Maintaining continuity is important to ensure grades are computed equitably. Therefore, while we may discuss exam solutions during office hours, we will not award additional points unless the regrade request form is completed and the appropriate channels are followed.

Key Dates

Mid-Term Exam	October 1 (in Lecture)
Resurrection Quiz	October 7 (in Discussion)
Last Day to Withdraw with a W	November 6
Mid-Term Exam	November 12 (in Lecture)
Resurrection Quiz	November 18 (in Discussion)
Final Exam	December 9 (2:00 - 5:00, Location TBD)

AI Use

If you would like to use AI tools for this course, you may use the AI-powered tool embedded within the Mastering Chemistry courseware as a personal study guide. This tool has been trained on validated chemistry content and designed for use in education that aligns with course content. If you use this tool to personalize your learning (answer questions, generate practice problems, provide summaries, etc), make sure you can solve problems independently. Note that using this AI tool or any other LLM to answer assigned homework problems is not a recommended usage of the tool, as doing so would be a disservice to building the skills and practice you need to perform on the exams, which represent the largest portion of your grade. AI tools may not be used to answer questions during class. Providing AI-generated answers to in-class questions (for example via Learning Catalytics) may be treated as a violation of the Duke Community Standard.

Honor Code

Students are expected to follow the honor code during exams, work individually, and use only the allowed materials: pens/pencils and a calculator without Wi-Fi capabilities. Communicating with others through watches or other means, using reference materials that are not allowed, or not handing in exams when time is called are all violations of the honor code and will result in an F in the course. After the first exam is handed out and before the last exam is collected, please do not communicate with anyone. Doing so violates the honor code, and we will report it to the student conduct board.

Chem Trivia – Just Some Fun!

1. Argentina is named after the element silver. The country's name is derived from argentum, the Latin word for silver, which is also the origin of the chemical symbol Ag. The name reflects historical legends of silver mountains and the "Río de la Plata" (River of Silver) region.

2. Apollo 13 (1995): The dramatic scene where engineers must fit a square lithium hydroxide (LiOH) canister into a round hole is based on historical fact; this chemical was crucial for removing from the air.
3. Mercury is the only liquid metal at room temperature, while tungsten has the highest melting point.
4. Many elements are named after places (e.g., Gallium for France, Germanium for Germany) or mythical figures.
5. While Calcium-40 is stable, our bodies contain tiny amounts of Radioactive Calcium-48.
6. The scientific name for "fool's gold" is iron pyrite (FeS_2).
7. Marie Curie (1867–1934) was a pioneering Polish French physicist who discovered polonium and radium, coined the term "radioactivity," and was the first person to win two Nobel Prizes (Physics 1903, Chemistry 1911). Her radioactive research notebooks are still so hazardous that they are stored in lead-lined boxes.
8. For reading this far, add your own chem trivia to the Ed Discussion Board to get +2 extra credit points on the first exam. I will add these at the end of the semester.

Accommodations

If you are a student with a disability and need accommodations for this class, it is your responsibility to register with the Student Disability Access Office (SDAO) and provide them with documentation of your disability. SDAO will work with you to determine what accommodations are appropriate for your situation. Accommodations are not retroactive, and I cannot provide disability accommodations until I receive a Faculty Accommodation Letter. Please get in touch with SDAO for more information at sdao@duke.edu or visit https://access.duke.edu/students/.

This class will use the Testing Center to provide testing accommodations to undergraduates who are registered with and approved by the Student Disability Access Office (SDAO) and to administer makeup tests for students with excused absences. The center operates by appointment only, and appointments must be made at least four (4) consecutive days in advance. However, please schedule your appointments as far in advance as possible.

If you have accommodations, you will not be able to make an appointment until you have submitted a Semester Request to the SDAO and your accommodations have been approved. If you have not already done so, promptly submit a Semester Request to the SDAO to ensure you can make your appointment on time. For instructions on how to register with SDAO, visit their website at https://access.duke.edu/students/.

For those taking make-up exams: The center handles make-up exams on a case-by-case basis. Please call the Testing Center at (919) 684 -1601 if you need to complete a make-up exam in less than four (4) days. You can book a make-up exam appointment as far in advance as possible using the center's website.

For instructions on making an appointment at the Testing Center, visit their website at https://testingcenter.duke.edu.

Religious Accommodations

University policy permits students to be absent from class to observe a religious holiday. Accordingly, Trinity College of Arts & Sciences and the Pratt School of Engineering have established procedures for students to notify their instructors of an absence necessitated by the observance of a religious holiday.

Please submit requests for religious accommodations at the beginning of the semester so we Can work to make suitable arrangements well in advance.

Behavior and Duke's Community Standard

Duke University is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflect upon and uphold these principles in all academic and non-academic endeavors and to protect and promote a culture of integrity.

Duke University has high expectations for students' scholarship and conduct. In accepting admission, students indicate their willingness to subscribe to and be governed by the rules and Regulations of the university, which flow from the Duke Community Standard (DCS).

Regardless of course delivery format, it is the responsibility of all students to understand and follow all Duke policies, including but not limited to the academic integrity policy (e.g., completing one's own work, following proper citation of sources, adhering to guidance around group work projects, and more). Ignoring these requirements is a violation of the DCS.

Students can direct any questions or concerns regarding academic integrity to the Office of Student Conduct and Community Standards at conduct@duke.edu and can access the DCS guide at https://dukecommunitystandard.students.duke.edu/.
A referral can be submitted at support.students.duke.edu.

Mental Health and Wellbeing Resources

Duke is committed to holistic student wellbeing, which includes one's mental, emotional, and physical health. The university offers resources to help students manage daily stress, to encourage intentional self-care, and to access just-in-time support. If you find you need support, your mental and/or emotional health concerns are impacting your day-to-day activities, your academic performance, or you need someone to talk to, the resources below are available to you:

A. DukeReach
DukeReach provides comprehensive outreach services to support students in managing all aspects of wellbeing, including referrals and follow-up services for students who are experiencing significant challenges related to mental health, physical health, social adjustment, and/or a variety of other stressors. You can contact the DukeReach team at dukereach@duke.edu and/or submit a referral at support.students.duke.edu.

B. Counseling and Psychological Services (CAPS)
CAPS offers counseling services to Duke students, including virtual appointments and referrals in the community. You do not need an appointment for an initial assessment. You may walk in or call 919-660-1000 to get started. Hours: Monday-Friday 9:00 am - 4:00 pm. After-hours counseling services are available at no additional cost to students. You can call: 919-660-1000 Option 2.

C. TimelyCareLinks to an external site. (formerly known as Blue Devils Care)
TimelyCare is an online platform that is a convenient, confidential, and free way for Duke students to receive 24/7 mental health support through TalkNow and scheduled counseling.

D. Duke Student Health
Student Health offers a wide range of healthcare services for all Duke students, many of which are covered by the student health fee. To make an appointment, call (919) 681-9355. Hours: Monday - Friday, 8 am - 4:30 pm, Thursday 9 am - 4:30 pm. Closed from 12-12:30 each day.

Title XI

Duke University is committed to encouraging and sustaining a learning, living, and work environment free from discrimination, harassment, and related misconduct in line with applicable laws and University policy.

Consistent with Title IX of the Education Amendment Act of 1972, Duke prohibits sex discrimination (including sexual harassment, sexual assault, dating and domestic violence, and stalking). Students who experience or are impacted by such conduct have the right to seek supportive measures, file a complaint, or pursue other options through Duke's Office for Institutional Equity (OIE).

All Duke faculty, other employees (including graduate students) with teaching or supervisory authority, Student Affairs professionals, HR representatives/managers, Athletics staff, and other Responsible Employees are required to share information about suspected discrimination, harassment, or related misconduct with OIE.

Academic Support Resources

The Academic Resource Center (the ARC) offers services to support students academically during their undergraduate careers at Duke. The ARC can provide support with time management, academic skills and strategies, course-specific tutoring, presentation skills, public speaking, and more. ARC services are available free to all Duke undergraduate students studying any discipline.

Learn more:
- Learning Consultations – time management, study strategies, learning preferences, and more
- VOICE Lab – strengthen public speaking, practice presentations, and more

You can contact the Academic Resource Center by phone at (919) 684-5917, by email at theARC@duke.edu, or by visiting http://arc.duke.edu/.

The TWP Writing Studio provides all writers in all disciplines with collaborative, writer-directed conversations at any stage in their processes. Writing consultants offer attentive, non-evaluative feedback to help you see whether you're expressing ideas in ways that are clear to someone outside our class, whether you're working on a writing assignment, oral presentation, poster, or other text-based project. Writing Studio consultations can be productive at any stage in your process, including before you have started writing. Visit http://twp.duke.edu/twp-writingstudio to learn more and to schedule an in-person or online appointment.

Teaching Objectives (These are objectives for me to cover)

Module 1: Nuclear and Atomic Structure

1.) Discuss the structure of the atom and nucleus.
2.) Explain binding energy and mass defects.
3.) Describe why nuclear decay occurs and the different pathways for nuclear decay.

4.) Complete calculations related to nuclear decay.
5.) Quantitatively compare the energy released from nuclear reactions to the energy released from chemical reactions.
6.) Review the key experiments that led to the understanding of atomic and nuclear structure.
7.) Explain the concept of isotopes and discuss how we quantify the relative amounts of each isotope.
8.) Explain the Bohr Model and illustrate key ideas that were used to develop equations for radii and energetics.
9.) Introduce the Schrödinger Equation and the concept of a wavefunction.
10.) Identify the components of a wavefunction and the shapes and energies of orbitals.
11.) Discuss how we use photoelectron spectroscopy to rationalize electron configurations.
11.) Qualitatively and quantitatively introduce the effective nuclear charge.
12.) Discuss trends in ionization energies, electron affinities, radii, and electronegativities.
13.) Briefly discuss the lanthanide contraction and the implications for 4d and 5d elements.
14.) Briefly discuss relativistic effects and properties of Au, Hg, Tl, and Pb.

Module 2: Solubility and Solutions

1.) Differentiate between intramolecular and intermolecular forces.
2.) Complete a thermodynamic cycle to explain why salts are soluble or insoluble.
3.) Review the concept of equilibrium and relate it to thermodynamics for the dissolution of salts.
4.) Define Ksp and review how to complete equilibrium calculations.
5.) Discuss the common ion effect and complete quantitative examples using the common ion effect.
6.) Identify the different types of colligative properties and complete related calculations.
7.) Explain how salt dissolution impacts the extent of the observed colligative properties.
8.) Define vapor pressure and complete quantitative calculations involving vapor pressure.
9.) Review Gas Laws as applied to vapor pressure calculations.

Module 3: Acid-Base Chemistry

1.) Review the autoionization of water.
2.) Discuss the pH scale and calculations with weak acids.
3.) Introduce mass and charge balances to solve for pH for very dilute acids.
4.) Solve for pH, pOH, etc. for weak acids (review from 101)
5.) Introduce calculations involving polyprotic acids.
6.) Qualitatively and quantitatively explain buffers.
7.) Introduce titrations to review acid-base calculations.
8.) Discuss the formation of metal hydrides and metal oxides.
9.) Use arrow-pushing formulations to describe the reactivity of metal and nonmetal oxides.
10.) Discuss the significance of solvent leveling (particularly in aqueous solutions).
11.) Discuss the role of carbon dioxide in promoting ocean acidification.
12.) Discuss Hard-Soft Acid-Base Theory and related applications.
13.) Use Molecular Orbital Theory to explain Hard-Soft Acid-Base Theory.

Module 4: Coordination Chemistry

1.) Discuss the structure of coordination complexes.
2.) Briefly outline the strategy for naming coordination complexes.
3.) Discuss isomerism with coordination complexes.
a. Structural (Linkage and Ionization Isomers)
b. Stereoisomers (Cis/Trans, Fac/Mer, Δ and Λ)
4.) Discuss UV/Vis Spectroscopy and Beer's Law to support experimental findings regarding the color of coordination complexes.

5.) Outline the tenets of crystal field theory and use these findings to explain the colors and magnetic properties of coordination complexes.
6.) Discuss the structure and function of cis-platin (an inorganic chemotherapy drug).
7.) Identify the associative and dissociative mechanisms for coordination complexes.
8.) Explain why coordination compounds are either labile or inert in chemical reactions.
9.) Differentiate between kinetic and thermodynamic control for mechanisms involving coordination complexes.
10.) Explain the concept of the trans effect and how it impacts the synthesis of coordination complexes.
11.) Discuss the Jahn-Teller effect and outline experimental evidence supporting this observation.
12.) Briefly introduce Ligand Field Theory and compare with Crystal Field Theory.
13.) Discuss the structure of Heme (from Hemoglobin).

Module 5: Redox Chemistry

1.) Explain the theoretical basis for redox chemistry.
2.) Outline the strategy for balancing redox reactions.
3.) Discuss the structure and function of voltaic and electrolytic cells.
4.) Outline strategies for completing quantitative calculations for standard and nonstandard conditions (the Nernst Equation).
5.) Explain how electrochemistry can be used to compute K_{sp} and K_a values.
6.) Discuss the organization and interpretation of Latimer and Pourbaix Diagrams.
7.) Explain how to predict the electrolysis products.
8.) Discuss challenges and achievements in the electrolysis of saltwater to produce hydrogen.
9.) Discuss the role of biochemical ion channels.
10.) Explain the significance of ferredoxins in biological pathways.

Module 6: Metallic and Ionic Bonding

1.) Explain how metallic bonds are formed.
2.) Identify crystal structures that are used to depict metallic compounds and relate to molecular orbital theory.
3.) Explain the concepts of band gap, conduction band, and valence band.
4.) Discuss the types of crystal structures observed for ionic compounds and define the concept of cubic, octahedral, and tetrahedral holes.

The 101/110 Learning Objectives

Some of this content will be reviewed as needed; however, all 101 instructors should have covered the objectives identified below. If not, please let me know. When teaching courses, especially the first in the sequence, the learning objectives must be covered to ensure students are prepared for the next course. Students with AP credit should have also seen each of the objectives below. The 101 textbook should have been Brown and LeMay or OpenStax 2.0.

Unit 1: Energy, Enthalpy, and Thermochemistry

LO1 Define and compare different types of energy, e.g. kinetic, potential, and thermal.
LO2 Define (and give examples or compare where relevant) the following terminology: universe, system, and surroundings; open, closed, and isolated systems; state of equilibrium; processes and changes of state; state function and path dependence/independence; extensive versus intensive properties; exothermic and endothermic.
LO3 Describe in words and in equations the first law of thermodynamics, in both the "global" (focused on changes in internal energy in the system, surroundings, and universe) and "system" points of view (focused on changes in internal energy of the system in relation to heat flow and work done on the

system).
LO4 Define heat and heat capacity, and perform calculations that interrelate heat, heat capacity, and temperature changes; define the specific heat capacity and the molar heat capacity and use them in calculations involving heat and temperature changes.
LO5 Apply the definition of P-V work to calculate the work done for constant pressure and constant volume processes; explain the sign convention choice for P-V work.
LO6 Describe and compare constant volume calorimetry and constant pressure calorimetry, including the experimental setups, how the heat of reaction is determined in a calorimetric experiment, and how the measured heat in each experiment relates to ΔE or ΔH.
LO7 Use the mathematical definition of the enthalpy H to relate ΔE and ΔH.
LO8 Explain what is meant by thermochemistry and how the first law of thermodynamics relates to chemical reactions.
LO9 Use the relation between ΔE and ΔH to determine one from the other for chemical reactions involving gases.
LO10 Explain how the value of ΔE or ΔH relates to the stoichiometry in a thermochemical equation, and more specifically, the effect on ΔE or ΔH of changing the stoichiometric coefficients by a constant factor or by reversing the chemical equation.
LO11 Define the standard enthalpy of formation, the related standard states, and what this definition implies as the reference energy choice for elements.
LO12 For a given chemical reaction, determine the enthalpy of reaction from standard enthalpies of formation.
LO13 Explain Hess's Law, and use it to determine the enthalpy change for a given reaction from the enthalpies of reaction for a series of other reactions involving the same chemical species.

Unit 2: Quantum Mechanics and Atomic Theory LO1 Define and compare properties associated with classical particles and waves, e.g., mass, velocity, momentum, energy, frequency, wavelength, intensity, speed, etc., and use the speed of light to relate frequency and wavelength for electromagnetic radiation.
LO2 Describe the main experimental results that led to the quantum theory of light and matter, including the spectrum of blackbody radiation, the photoelectric effect, and the line emission spectra of atoms.
LO3 Discuss how these experiments led to the various hypotheses of Planck, Einstein, Bohr, and deBroglie regarding the wave nature of matter, and use the equations they proposed to carry out calculations relating energy to frequency, momentum to wavelength, and the energies of Bohr orbits in hydrogen-like atoms to line spectra and ionization energies.
LO4 Discuss the physical interpretation and meaning of the solutions to the Schroedinger equation, in particular the wavefunction as a probability amplitude and the square of the wavefunction as a probability density.
LO5 Explain how the wave functions or atomic orbitals that result from solving the Schroedinger equation for hydrogen-like atoms relate to probability amplitudes, probability densities, and boundary surface plots for electrons.
LO6 Explain what aspects of the electron orbitals and other properties are governed by each of the four orbital quantum numbers n, l, ml, and ms, and be able to list the possible values for each.
LO7 Describe the shapes of orbitals for a given set of quantum numbers through $n = 3$.
LO8 Distinguish between energy levels and energy states or orbitals, and determine the number of degenerate orbitals .
LO9 Determine electron configurations and orbital diagrams for the ground electronic states of atoms in the periodic table through period 4 by applying the Aufbau Principle along with the Pauli exclusion principle and Hund's rule, and relate these electron configurations to the position of an atom in the

periodic table.
LO10 Use the Pauli principle and Hund's rule to determine the number of unpaired electrons in an atom, and hence whether the atom is paramagnetic or diamagnetic; know the exceptions to the usual orbital diagrams that occur for Cr and Cu.
LO11 Describe how trends in n and Zeff across a period and down a group in the periodic table affect orbital size and stability, and thus trends in periodic properties of atoms .
LO12 Define the following atomic properties: atomic radius, ionization energy, and electron affinity; describe the periodic trends in each of these properties and how they relate to the underlying trends in n and Zeff; and finally, be able to rank a series of atoms according to the values of these properties.
LO13 Explain how basic chemical properties of atoms, such as whether they are metallic or non-metallic, or the reactivity of a series of metals in water, can be explained in terms of their electronic configurations and the underlying trends in electronic configurations and orbital properties .

Unit 3: Bonding: General Concepts
LO1 Describe the bonding that occurs in ionic compounds and in covalent and polar covalent bonds, and compare them.
LO2 Write proper Lewis dot structures for atoms, explain the "octet rule," and apply it to molecules and compounds.
LO3 Write the ground state electron configurations for ions, use periodic trends to rank a series of ions according to their size, including an isoelectronic series of ions.
LO4 Explain which atoms in the periodic table are likely to form ionic compounds and why, and use Lewis structures and the octet rule to predict the formulas of ionic compounds.
LO5 Define the lattice energy and describe its dependence on ionic charges and the distance between them.
LO6 Use Hess's Law in the form of a Born-Haber cycle to interrelate the lattice energy, ionization energies, electron affinities and other relevant enthalpy changes (such as enthalpies of formation or bond dissociation energies), and be able to deduce the value for one of these quantities given values for the others.
LO7 Explain the stabilities and formulas of ionic compounds in terms of the trade-off between lattice energy and ionization energies, for example, $MgCl_2$ versus MgCl.
LO8 Draw proper Lewis dot structures for covalently bonded molecules, including those with lone pairs, multiple bonds, and overall ionic charges.
LO9 Describe the trends in bond length and bond energies with bond multiplicities.
LO10 Use bond energies to estimate enthalpies of reaction.
LO11 Define electronegativity, use the periodic trends in electronegativity to rank a series of atoms, and use electronegativities to determine whether a given bond will be ionic, polar covalent, or non-polar covalent.
LO12 Deduce the formal charges in a Lewis structure, and use them to rank different possible structures according to their stability or plausibility.
LO13 Know which types of atoms can form molecules with octet rule exceptions, including expanded octets, incomplete octets, and odd electron molecules; draw Lewis structures for such molecules.
LO14 Draw a series of resonance structures for molecules where they are possible, and use these structures and their formal charges to predict properties of the molecules, such as bond lengths or bond multiplicities.
LO15 Starting from a molecular formula or Lewis structure, use the VSEPR model to deduce electron geometries and molecular geometries for systems with two to six terminal atoms around a central atom.
LO16 Describe how lone pairs distort molecular geometries within the VSEPR approach and predict the bond angles for a related series of molecules (e.g., CH_4 versus NH_3 versus H_2O).
LO17 Define the electric dipole moment, and use electronegativities to predict the relative polarities of

bonds, and the directions in which the dipole "arrow symbols" should point.
LO18 Use the bond dipole model with molecular geometries to determine whether a given molecule will be polar or non-polar.

Unit 4: Covalent Bonding: Orbitals
LO1 Describe how covalent bonding relates to the overlap or constructive interference of atomic orbitals, and to electron density between the bonded atoms.
LO2 Sketch a typical graph illustrating how the energy of a diatomic molecule varies with the internuclear distance R, explain which interactions result in the minimum in energy and the sharp rise in energy at small R, and identify on the graph the molecular "equilibrium" bond length and dissociation energy.
LO3 Describe how bonds are formed in Valence Bond Theory by overlapping half-filled atomic orbitals on neighboring atoms, and draw pictures to illustrate this for simple cases like HF or F_2.
LO4 Describe the rationale for introducing atomic orbital hybridization, as well as the conceptual process involved in forming hybrid orbitals (e.g., electron promotion, hybridization, bonding, and the balance of energy cost and gain).
LO5 Sketch hybrid orbitals for sp, sp^2, sp^3, dsp^3, and d^2sp^3, and describe the geometry of the hybrid orbitals around an atom for these cases (e.g., linear, tetrahedral, etc).
LO6 For a given molecule, be able to go from the Lewis structure, to the electron geometry, hybridization, and bond angles around each atom, to a sketch of the valence bonding orbitals for the entire molecule, including both sigma and pi bonds.
LO7 Use the sigma and pi bonding in a given molecular structure to deduce the overall geometry of the molecule, especially in cases involving multiple bonds (e.g., the planarity of ethene).
LO8 Describe how approximate molecular orbitals in the MO approach are formed from linear combinations of atomic orbitals.
LO9 For diatomic molecules, be able to describe and sketch the sigma and pi bonding and anti-bonding orbitals that result from combinations of the s and p orbitals on the two atoms.
LO10 Use molecular orbital energy level diagrams to show how the electrons in a homonuclear diatomic molecule occupy the various molecular orbitals, and from this diagram determine the bond order, MO electron configuration, and the number of unpaired electron spins (hence paramagnetic vs diamagnetic); also be able to go from a MO electron configuration to MO diagram.
LO11 For a series of diatomic molecules, use bond orders to determine their relative stabilities, bond lengths, and bond energies.
LO12 Extend the use of molecular energy level diagrams described above to heteronuclear diatomic molecules and diatomic molecular ions.
LO13 Describe how the molecular orbital approach can describe delocalized bonding, for example in hydrocarbons with conjugated pi systems, and thereby explain the averaging associated with resonance structures in the Lewis model of bonding.

Unit 5: Chemical Kinetics
LO1 Explain what is meant by rate of reaction or reaction rate.
LO2 Explain how reaction rates are expressed and their relation to stoichiometry.
LO3 Describe experimental methods that have been used to measure reaction rates.
LO4 Explain what rate laws are and how they are determined.
LO5 Use experimental data to derive rate laws for chemical reactions.
LO6 Compare and contrast integrated rate laws for zero, first, and second-order reactions.
LO7 Identify factors that influence the rate of reaction and how each factor affects the rate.
LO8 From a molecular perspective, explain why the reaction rate is affected when conditions are changed, including how the collision model can provide a molecular explanation for the dependence

of a second-order reaction on concentration and temperature.
LO9 Use equations associated with chemical kinetics to determine reaction rates, half-lives, rate constants, reaction orders, activation energies, and reactant concentrations versus time.
LO10 Describe the components of reaction mechanisms, including elementary steps, molecularity, intermediates, and catalysts.
LO11 Evaluate reaction mechanisms as to their feasibility.
LO12 Derive rate laws from reaction mechanisms using common approximations, including the rapid pre-equilibrium/rate-limiting-step approximation.
LO13 Apply the principles of chemical kinetics to various reactions, such as those associated with environmental chemistry.

Unit 6: Chemical Equilibrium LO1 Describe the equilibrium condition from macroscopic and molecular perspectives.
LO2 Describe the changes in concentrations of the reactants and products as a chemical system moves towards a state of equilibrium from a given set of initial conditions.
LO3 Describe the changes in the forward and reverse reaction rates as a chemical system moves towards a state of equilibrium from a given set of initial conditions.
LO4 Relate the equilibrium constant to the rate constants for the forward and reverse reactions.
LO5 Describe how the law of mass action relates to chemical equilibrium.
LO6 Explain how there can be an infinite number of initial conditions that result in an infinite number of equilibrium positions, yet only one value for the equilibrium constant for a reaction at a fixed temperature.
LO7 Define the reaction quotient Q, relate Q to the equilibrium constant K, and explain how Q can be used to predict whether a system is at equilibrium, and if not, the direction in which a chemical system will move to establish equilibrium.
LO8 Given a chemical equation, write the expression for the reaction quotient and/or equilibrium constant.
LO9 Interpret the magnitudes of equilibrium constants regarding the extent of reaction when equilibrium has been established.
LO10 Distinguish between K_c and K_P and explain how they are related.
LO11 Manipulate chemical equations with known equilibrium constants to predict the equilibrium constant for another reaction.
LO12 Perform quantitative calculations involving Kc, KP, and concentrations or pressures, e.g., determine the value of K_c from equilibrium concentrations, or from the changes to initial concentrations that hold at equilibrium; use Kc plus initial concentrations to determine final equilibrium concentrations (using ICE charts).
LO13 Identify factors that can stress a system in a state of equilibrium and explain why these factors stress the equilibrium system.
LO14 Of the above factors, identify those that will alter the magnitude of the equilibrium constant, and how the constant will change as a result of the stress.
LO15 Apply Le Châtelier's principle to predict the response of an equilibrium system to a stress.

Unit 7: Acids & Bases
LO1 Characterize acidic and basic solutions based on their macroscopic properties.
LO2 Characterize compounds as acids and bases according to the Arrhenius definition.
LO3 Identify common strong acids and bases by name and formula.
LO4 Characterize compounds as acids and bases according to the Brønsted-Lowry definition.
LO5 Identify Bronsted-Lowry Acid-Base pairs (acid and conjugate base, base and conjugate acid) in a given reaction.

LO6 Characterize compounds as acids and bases according to the Lewis definition.
LO7 Identify factors that influence the strength of a substance as an acid or as a base and use them to predict relative strengths of a given group of substances.
LO8 Describe the relationship between the strengths of acids and their conjugate bases.
LO9 Write equations and equilibrium expressions for the ionization or dissociation of an acid and for the ionization or dissociation of a base in water.
LO10 Write the equation for the auto-ionization of water and relate to $K_w, [H^+], [OH^-]$, pH, and pOH.
LO11 Describe changes in Kw, [H+], [OH-], pH, and pOH as a solution increases in acidity or in alkalinity
LO12 Characterize acids and bases in terms of their strength using the following properties:
$K_a, pK_a, K_b, pK_b, [H^+], pH, [OH^-], pOH$, percent ionization in water, and strength relative to water.
LO13 Use $K_a, pK_a, K_b, orpK_b$ to predict the extent of an acid-base reaction and the value of the equilibrium constant for the reaction.
LO14 Perform quantitative equilibrium calculations related to water solutions of acids, bases, and salts ($K_a, pK_a, K_b, pK_b, [H^+], pH, [OH^-], pOH$, and percent ionization), including polyprotic acids.

Unit 8: Spontaneity, Entropy, and Free Energy
LO1 Describe what is meant by a “spontaneous process” and provide simple examples.
LO2 Describe in words and equations the second law of thermodynamics and the entropy S.
LO3 Describe the third law of thermodynamics and how this leads to absolute entropies.
LO4 Predict trends in absolute entropies among a series of substances, for example, with respect to physical state, molecular complexity, bond strength, and molecular mass.
LO5 Calculate the entropy change for phase changes or for chemical reactions from absolute entropies.
LO6 Apply the second law to processes like phase transitions to determine whether they are spontaneous or not, or to find the transition temperature where two phases are in equilibrium.
LO7 Define the Gibbs Free energy and its relation to ΔS and ΔH.
LO8 Determine the standard Gibbs Free Energy for a reaction or other process from standard Gibbs Free Energies of formation, or from a combination of standard enthalpies of formation and absolute entropies.
LO9 Relate Q and K to the thermodynamic quantities ΔG, ΔG^o, ΔH^o, and ΔS^o for a reaction, and use the thermodynamic quantities to determine the temperature dependence of K.
LO10 Describe the relationship between entropy and “disorder,” “randomness,” and “probability” as embodied in the Boltzmann equation.